

STUDENT ENROLLMENT & COURSE MANAGEMENT SYSTEM

6. PROJECT TESTING

6.1 Testing Objective

The objective of testing is to ensure that the Student Enrollment & Course Management System functions correctly, maintains accurate student and course records, provides secure access, and supports smooth student enrollment. Testing helps identify errors, verify system performance, and ensure reliability and usability.

6.2 Test Areas

Test ID	Area	Expected Result	Status
TC-01	Student Record Creation	Student details can be created and stored correctly.	Tested
TC-02	Course Record Creation	Course information is saved correctly.	Tested
TC-03	Faculty Record Creation	Faculty details can be created and maintained.	Tested
TC-04	Enrollment Record Creation	Student enrollment details are stored correctly.	Tested
TC-05	Mandatory Field Validation	Required fields prevent incomplete records.	Tested
TC-06	Email Validation	Invalid email formats are rejected.	Tested
TC-07	Enrollment Validation	Enrollment records are linked to the correct student and course.	Tested
TC-08	Course Availability	Students can view available courses correctly.	Tested
TC-09	Relationship Testing	Student, course, faculty, and enrollment relationships display correctly.	Tested
TC-10	Security Testing	User roles provide appropriate access permissions.	Tested
TC-11	Access Control	Users can access only authorized records and features.	Tested
TC-12	Reports	Reports display accurate student and enrollment information.	Tested
TC-13	Dashboards	Dashboard components reflect available enrollment data.	Tested
TC-14	Performance	Pages and reports are checked against the target of approximately less than 3 seconds.	Checked

6.3 Validation Testing

Validation testing ensures that student, course, and enrollment details are entered correctly. It checks mandatory fields, email formats, valid data, and enrollment relationships to prevent incomplete or incorrect records.

6.4 Security and Access Testing

Security testing ensures that students, faculty, and administrators can access only their authorized features and records. It verifies secure login, role-based permissions, and protection of student and course information

6.5 Analytics Testing

Analytics testing ensures that student enrollment reports and dashboards display accurate and updated information. It verifies enrollment counts, course details, and report generation to support effective academic management.

6.6 Testing Conclusion

Testing confirms that the Student Enrollment & Course Management System performs its major functions correctly. Student registration, course management, enrollment processing, data validation, security, and report generation are checked to ensure accuracy, reliability, and efficient system operation.